Operator

- selects bend sides
- defines wall heightbasic / expert mode

Embedded 3D Sheet Bending Viewer

- interactive STL mesh visualisation
- visual bend-side selection
- user-guided input
- runs as local web service in drag&bot
- viewer terminates after confirmation.

STL File Input

- workpiece STL model

Bend Selection Module
Separate selected bends into:

- primary-side bend list
- opposite-side bend list

Parameter Processing Module

- validate user selections
- process wall height
- process mode-specific inputs
- generate structured parameter set.

Geometry Processing Module

- sheet geometry extraction
- hole detection
- feature classification

Primary Bending Module

- Receives primary side bend list

Opposite-Face Workflow Module

- receives opposite-side bend list

Grip Planning Input Module

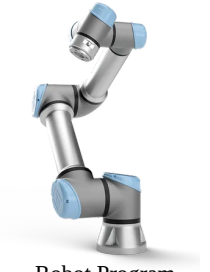
- receives geometry data
- receives grip-related parameters
- receives sheet mass and dimensions
- receives TCP / tool-related input if
- needed for grasp setup

Safety Validation Input Module

- bend selection consistency check
- execution authorisation input
- pre-execution input validation

Planning / Trajectory Input Module

- receives task configuration
- parameters
- receives bend sequence
- information
- receives structured planning input



Robot Program Generation / Execution